

# Performance Metrics for Clustering

## 1. Introduction

Clustering evaluation differs fundamentally from regression and classification because, in most cases, there are no ground-truth labels. Therefore, clustering metrics aim to measure:

- **Compactness:** How close data points are within a cluster
- **Separation:** How far apart clusters are from each other
- **Agreement:** How well clustering matches true labels (if available)

These metrics are broadly divided into:

- **Internal Metrics:** Do not require ground truth labels
- **External Metrics:** Require ground truth labels

## 2. Silhouette Score

### Definition

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \quad (1)$$

### Explanation

- $a(i)$ : Average distance of point  $i$  to points in the same cluster
- $b(i)$ : Minimum average distance of point  $i$  to points in other clusters

### Interpretation

- $s(i) \approx 1$ : Well clustered
- $s(i) \approx 0$ : On boundary
- $s(i) < 0$ : Misclassified

### Insight

Combines cohesion and separation in a single metric.

### 3. Rand Index (RI)

#### Definition

$$RI = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

#### Explanation

Measures similarity between predicted clustering and true labels using pairwise agreements.

#### Limitation

Does not account for agreement due to chance.

### 4. Adjusted Rand Index (ARI)

#### Definition

$$ARI = \frac{RI - \text{Expected RI}}{\text{Max RI} - \text{Expected RI}} \quad (3)$$

#### Interpretation

- 1: Perfect clustering
- 0: Random clustering
- $< 0$ : Worse than random

#### Insight

Corrects Rand Index for random chance.

### 5. Mutual Information (MI)

#### Definition

$$MI(U, V) = \sum_{i,j} P(i, j) \log \left( \frac{P(i, j)}{P(i)P(j)} \right) \quad (4)$$

#### Explanation

Measures how much information is shared between true labels and predicted clusters.

#### Limitation

Not normalized, making comparisons difficult across datasets.

## 6. Calinski-Harabasz Index

### Definition

$$CH = \frac{\text{Between-cluster variance}}{\text{Within-cluster variance}} \quad (5)$$

### Explanation

Evaluates clustering based on ratio of separation to compactness.

### Interpretation

Higher values indicate better clustering.

## 7. Davies-Bouldin Index

### Definition

$$DB = \frac{1}{k} \sum_{i=1}^k \max_{j \neq i} \left( \frac{\sigma_i + \sigma_j}{d(c_i, c_j)} \right) \quad (6)$$

### Explanation

- $\sigma_i$ : Intra-cluster distance
- $d(c_i, c_j)$ : Distance between cluster centers

### Interpretation

Lower values indicate better clustering.

## 8. Comparison of Metrics

| Metric              | Type     | Measures                   | Ideal Value |
|---------------------|----------|----------------------------|-------------|
| Silhouette Score    | Internal | Cohesion + Separation      | Close to 1  |
| Rand Index          | External | Pairwise Agreement         | Close to 1  |
| Adjusted Rand Index | External | Chance-corrected Agreement | Close to 1  |
| Mutual Information  | External | Shared Information         | High        |
| Calinski-Harabasz   | Internal | Variance Ratio             | High        |
| Davies-Bouldin      | Internal | Cluster Similarity         | Low         |

## 9. Conclusion

Clustering metrics provide different perspectives:

- **Silhouette**: Individual point quality
- **Rand / ARI**: Pairwise agreement

- **Mutual Information:** Information theory perspective
- **CH Index:** Variance-based separation
- **DB Index:** Cluster overlap

Selecting the right metric depends on whether ground truth labels are available and the structure of the data.